# Younsoo Park

danielpark0605@gmail.com yqp5187@psu.edu +1 (814) 810-1707 GitHub: danielsoo

## Education

The Pennsylvania State University University Park, PA

Bachelor of Science in Computer Science Minor in Mathematics Expected Graduation: May 2027
Dean's List

## Research Interests

Systems Security
Network Security
Distributed Systems
LLM Systems
AI Agent Memory
Federated Learning
TinyML for IoT Security
Adversarial Machine Learning
Edge AI Systems

## Research Experience

Undergraduate Research Assistant & Team Lead Oct 2025 – Present

Reliable Federated TinyML Deployment for IoT Security (WIP, LCTES '26)

Penn State University and Elizabethtown College Supervisors: Dr. Suman Saha and Dr. Peilong Li

Research focus: Federated and adversarially robust TinyML systems for IoT security and resource-constrained edge-device deployment.

Key contributions:

- Designed a federated learning pipeline using FedAvgM, focal loss, and cosine learning-rate scheduling on CIC-IDS2017.
- Improved attack recall from 46.7% to 93.85% and F1-score from 84.1% to 89.32%.
- Built a TinyML compression pipeline including BatchNorm folding, structured pruning, knowledge distillation, and INT8 QAT/PTQ.
- Achieved 12.28× model compression and 74.5% inference latency reduction for ESP32 deployment.
- Evaluated adversarial robustness using PGD attacks ($\varepsilon$-sweep 0.01–0.2) across 48 compression configurations.

## Publications and Manuscripts

Reliable Federated TinyML Deployment for IoT Security Work in Progress Target Venue: ACM LCTES 2026

## Independent Study

AI Agent Memory Systems — Jul 2026 – Present

Advisor: Prof. Kiwan Maeng

- Researched five production and research AI agent memory systems (Mem0, A-MEM, LangMem, Zep, OpenAI Memory) to identify the SOTA approach, comparing memory creation, retrieval, and lifecycle-management approaches.
- Identified a core architectural divergence: Mem0 extracts and retrieves discrete facts, while A-MEM models memory as an evolving, interconnected knowledge structure; traced how Mem0's open-source implementation diverged from its published ADD/UPDATE/DELETE design into an ADD-only, retrieval-time resolution approach.
- Narrowed research focus to Mem0 (implementation depth, practical extensibility) and A-MEM (architectural novelty) for application to the lab's production AI agent system.
- Applying memory architectures to production recommendation systems through a Memory-first Personalization framework, shipped as Wowness at Levit Inc.

## Industry and Technical Experience

Levit Inc. (Shopport AI) — Seoul, South Korea

Associate Problem Solver — June 2026 – Aug 2026

- Worked directly with the CEO to define product direction, prioritize engineering initiatives, and translate customer problems into production AI systems for an LLM-powered shopping assistant.
- Built a production evaluation platform analyzing 49,807 conversations and 80,124 query-level routing decisions, achieving 97.7% evaluation coverage through versioned routing audit and historical backfill pipelines that reprocessed 47,231 conversations spanning 100+ days of production history.
- Developed end-to-end experimentation infrastructure across frontend, backend, workers, and admin systems, improving recommendation completion from 62.8% to 75.0% (+12.2 percentage points).
- Evolved Shopport's routing stack through three architectural generations – LLM classification, an embedding-based semantic fast path ($\sim 8\times$ faster for confident cases), and a hybrid classifier for ambiguous queries – reaching 86% V3 routing accuracy and rescuing 93% of eligible router errors.
- Designed and shipped Wowness, a Mem0-backed cross-category discovery recommender grounded in the live catalog, as sole engineer across a 919-line NestJS engine, React UI, and admin review lab – extending the Memory-first Personalization architecture from proposal to production.
- Diagnosed production failures at the system boundary, including a split MongoDB connection that held the V3 feature switch at zero executions, and corrected systematic analytics false positives, reducing one queue-level count from 2,005 to 811.
- Contributed 730 GitHub commits and 143 merged pull requests, adding over 114,000 lines of TypeScript/TSX during the engagement.

SIGNUM (Nittany AI Alliance – PIT-UN Funded) — Aug 2024 – Nov 2025

Co-founder of a $3,750 PIT-UN funded AI platform for caregiver decision support.

- Built multi-source ETL pipelines integrating CMS, NPPES, and Google Places datasets into a DuckDB warehouse.

- Developed hospital-rating prediction models using Markov Transition analysis with confidence intervals.
- Designed and deployed domain-specific RAG systems using FAISS and AWS Bedrock, reducing caregiver research time from 50+ hours per week to near-instant answers.

Atom Tech Solutions LTD — Berkeley, CA
Software Engineering Intern — May 2022 – Aug 2022

- Built secure authentication systems using SQL/JWT authentication and encrypted credential storage.
- Implemented automated credential generation and authenticated onboarding workflows.

## Technical Projects

GladIEEEators – Penn State IEEE Battle Bots (1st Place) — Apr 2026

- Designed Shot & Chaser, a coordinated two-robot combat system, for the IEEE Student Chapter Battle Bots Competition, winning 1st place.
- Engineered a belt-driven AR500 vertical-spinner weapon system (334 J at 18,500 RPM, 200+ mph tip speed), validated via Fusion FEA above 4,000 N with a 2.09 minimum safety factor.
- Designed mixed-material armor (6061 aluminum, UHMW, TPU) and integrated drivetrain, radio control, and protected LiPo electronics across both robots within a $976 total build budget.

XIIO – LinkedIn for Filmmakers — Apr 2026 – Present

- Built a creator networking platform featuring portfolios, video streaming, collaboration workflows, and school-based discovery using Next.js, Firebase, and Cloudflare Stream.
- Designed real-time messaging, creator profiles, media uploads, and production portfolio management.

ASME @ Penn State Website — Aug 2024 – Present

- Developed the official ASME Penn State website using React, TypeScript, Vite, and Firebase.
- Designed a multi-tier permission and approval system for organization management.

Hangukgwan Restaurant, Taiwan, Software Engineer — Summer 2022, Summer 2025

- Built a four-language (Korean, English, Simplified and Traditional Chinese) ordering and information site for a two-location restaurant in Zhubei, Taiwan, using Next.js, React, TypeScript, Tailwind CSS, and MongoDB Atlas, with menu, takeout, and account pages.
- Implemented JWT authentication and Google Maps geocoding for the restaurant's two locations.
- Built an Android POS tablet app in Java that prints new orders directly to the kitchen printer over TCP (port 9100), replacing a Chrome-based flow that Android blocks for unattended printing; prepared the Google Play internal-testing release.

## Teaching Experience

Lead Teaching Assistant, Math 230 (Multivariable Calculus) Penn State University Aug 2025 – Present

Responsibilities include assisting instruction, supporting students during office hours, and helping coordinate course activities.

## Leadership and Service

THON Technology Committee Technology Captain (2027) Apr 2026 – Present

Lead year-round technology operations for THON, which raises $10M+ annually: maintain the organization's primary website, ship new features, and oversee the technical systems the organization relies on.

Penn State Advanced Vehicle Team (AVT) Project Manager Fall 2026 – Present

Manage planning and deliverables for Penn State's team in Year 1 of the EcoCAR Challenge (managed by Argonne National Laboratory; sponsored by GM and MathWorks), building a fully electric vehicle while establishing the structure and processes that later teams will build on.

Association for Computing Machinery (ACM) Web & Design Director Apr 2026 – Present

Republic of Korea Air Force Squad Leader Sept 2022 – June 2024

## Relevant Coursework

Computer Science

- CMPSC 131: Programming and Computation I
- CMPSC 132: Programming and Computation II
- CMPSC 221: Object Oriented Programming
- CMPSC 311: Introduction to Systems Programming
- CMPSC 360: Discrete Mathematics
- CMPSC 461: Programming Language Concepts
- CMPSC 464: Theory of Computation
- CMPSC 465: Data Structures and Algorithms

Computer Engineering

- CMPEN 270: Digital Design
- CMPEN 331: Computer Organization and Design

## Awards and Honors

1st Place, ThinkNeuro LLC Project Competition Apr 2026

1st Place, IEEE Student Chapter Battle Bots Competition Apr 2026

PIT-UN Funding Award ($3,750) Apr 2025

3rd Place, ACM Pathfinder Robotics Challenge Apr 2025

Republic of Korea Air Force Commendation Award Aug 2023

## Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL

Frameworks and Tools: React, Node.js, NestJS, Express, Flask, MongoDB, Redis, Firebase, Kubernetes, Git, AWS Bedrock, DuckDB

Machine Learning: TensorFlow, PyTorch, FAISS, Mem0, SHAP, spaCy, Scikit-learn, Flower